

Nonlinear Semiclassical Spectroscopy of Ultrafast Molecular Polariton Dynamics

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We introduce a theoretical framework that allows for the systematic and efficient description of the ultrafast nonlinear response of molecular polaritons, i.e., hybrid light-matter states, in the collective regime of large numbers of molecules $\mathcal{N}$ coupled to the cavity photon mode. Our approach is based on a semiclassical, mean-field evolution of the molecular Hamiltonian and the cavity field, which is complemented by a perturbative expansion of both light and matter counterparts in the input fields entering the cavity. In addition, expansion in terms of the pulse phases enables us to disentangle different excitation pathways in Liouville space, thereby distinguishing contributions to the nonlinear response. The formalism extends traditional free-space nonlinear spectroscopy by incorporating the feedback of matter onto the light field via the induced polarization. We demonstrate the utility of the framework by applying it to the calculation of pump-probe polariton spectra and show how, by storing the pulses, the cavity facilitates additional excitation pathways, which can be used to isolate purely bright state contributions. Our method, which does not scale with $\mathcal{N}$, is broadly applicable and can be extended to model a wide range of current experiments investigating the dynamical nonlinear response of hybrid light-matter states.

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Introduction—Molecular polaritons, quasiparticles resulting from the strong coupling between confined photonic modes and vibrational or electronic excitations of molecules, e.g., inside Fabry-Pérot cavities, have gained significant attention over recent years, largely due to their potential applications in areas such as polariton chemistry [1,2], energy transport [3,4], condensation and lasing [5,6], and the manipulation of optical nonlinearities [7–10]. Recent advances in electronic [11–19] and vibrational [8,20–25] nonlinear polariton spectroscopy have provided new insight into the nonlinear response of molecular polaritons by revealing the dynamics of the complex interactions and energy relaxation pathways. However, significant challenges and open questions remain in understanding the nonlinear polariton response, particularly regarding the short-time dynamical behavior, i.e., before the cavity photons, and thus the polariton states, have decayed ($\sim$ fs–ps in experiments). The development of an efficient formalism that can address these questions is crucial for advancing the understanding of polariton dynamics across different time scales as observed in pump-probe experiments.

While the theoretical formalism of nonlinear spectroscopy involving the interaction between laser light and matter in free space is well established [26–31], recent works have aimed at constructing a framework for the theory of nonlinear electronic and vibrational polariton spectroscopy [32–37]. Some studies have considered direct

excitation of the molecules through the side of the cavity [34], whereas other works have taken a fully quantum mechanical approach, thereby restricting the scope of the treatment to either long times after light and matter have decoupled, or the explicit simulation of only a few molecules [33,36].

In this Letter, we introduce a systematic formalism to nonlinear cavity spectroscopy in the limit of large numbers of molecules $\mathcal{N}$ per photon mode. Our approach is based on a mean-field evolution of the coupled light-matter system [38,39], related to semiclassical Maxwell-Liouville approaches which have been developed, e.g., in semiconductor optics and nanophotonics [40–48]. We perturbatively expand both the cavity field and density matrix in the input fields driving the cavity and show that the formalism can be regarded as an extension of the conventional free-space perturbative nonlinear spectroscopy framework by Mukamel [26], now incorporating nonlinear contributions to the field resulting from the feedback of the molecules onto the cavity dynamics. Expansion of the nonlinear response in terms of phase components allows us to separate different excitation pathways in Liouville space and identify purely bright state contributions, using the widely adopted toolkit of double-sided Feynman diagrams.

Mean-field approach—We consider an ensemble of $\mathcal{N}$ molecules collectively coupled to the single confined mode of an optical cavity at frequency ω_c [see schematics in Fig. 1(a)]. Each molecule j is described by a (in principle arbitrary) Hamiltonian $\mathcal{H}_0^j$ with a corresponding dipole operator $\hat{\mu}_j$. The interaction between the molecules and the cavity field is described by a Dicke-type Hamiltonian such

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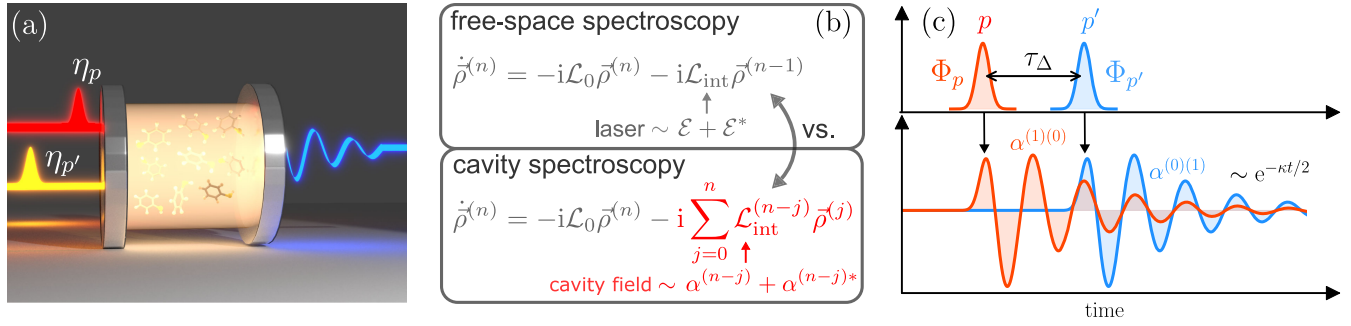


FIG. 1. (a) Schematics. An optical cavity containing an ensemble of $\mathcal{N}$ molecules is driven by input pulses with amplitudes η_p (pump) and $\eta_{p'}$ (probe). (b) Key distinction between conventional, free-space nonlinear spectroscopy and nonlinear cavity spectroscopy, concerning the evolution of the molecular density matrix. For cavity spectroscopy, the feedback of the molecular polarization onto the cavity field leads to a coupling of the n th order molecular density matrix $\bar{\rho}^{(n)}$ to all lower orders in the cavity field, as opposed to the free-space case, where the interaction is always linear in the laser field that drives the molecules. (c) Pump and probe pulses entering the cavity at time delay τ_Δ get stored and prolonged in the cavity on a time scale given by the cavity lifetime κ^{-1} , thereby allowing the (linear) cavity field created by the probe ($\alpha^{(0)(1)}$) to act before the pump ($\alpha^{(1)(0)}$). Additional phases $\Phi_{p,p'}$ imprinted on the pulses can be utilized to separate excitation pathways in Liouville space.

that the total Hamiltonian describing the full quantum evolution of the coupled light-matter system is expressed as (we set $\hbar \equiv 1$)

$$\mathcal{H} = \omega_c \hat{a}^\dagger \hat{a} + \sum_{j=1}^{\mathcal{N}} \mathcal{H}_0^j + E_0(\hat{a} + \hat{a}^\dagger) \sum_{j=1}^{\mathcal{N}} \hat{\mu}_j, \quad (1)$$

with the cavity zero-point amplitude $E_0 = \sqrt{\omega_c/(2\epsilon_0\mathcal{V})}$, where $\mathcal{V}$ is the cavity mode volume and ϵ_0 is the vacuum permittivity. To facilitate the understanding of the formalism, we initially assume that the cavity field is driven by only a single input field with amplitude η at a central carrier frequency ω_ℓ and with a Gaussian pulse shape described by the temporal envelope function $f(t)$ with a temporal pulse width τ_w .

The limit of large $\mathcal{N}$ allows us to apply a mean-field approximation (exact for $\mathcal{N} \rightarrow \infty$), which amounts to setting $\langle \hat{a} \rangle = \alpha(t)$ [38]. Additionally, for illustration, we take all molecules to be identical, i.e., $\mathcal{H}_0^j = \mathcal{H}_0$, $\hat{\mu}_j = \hat{\mu}$, although extension to disordered ensembles offers no conceptual difficulty. The simplified mean-field Hamiltonian then expresses as

$$\mathcal{H}_{\text{MF}} = \mathcal{H}_0 + E_0[\alpha(t) + \alpha^*(t)]\hat{\mu}. \quad (2)$$

The dynamical evolution of the (now single-molecule) density matrix ρ is governed by the master equation $\partial_t \rho = i[\rho, \mathcal{H}_{\text{MF}}] + \mathcal{D}[\rho]$. Here, the dissipator $\mathcal{D}[\rho]$ represents possible additional loss channels of the molecule (e.g., dephasing at rate γ_ϕ , nonradiative decay, etc.) that may be added, e.g., in Lindblad form. However, a complete description of the mean-field problem also involves solving the classical equation for the cavity field amplitude which undergoes damping at rate κ (photon loss)

$$\dot{\alpha}(t) = -\left(\frac{\kappa}{2} + i\omega_c\right)\alpha(t) - i\mathcal{N}E_0P(t) - \eta f(t)e^{-i\omega_\ell t}, \quad (3)$$

containing the induced molecular polarization $P(t) = \text{Tr}[\hat{\mu}\rho(t)]$ of all $\mathcal{N}$ molecules. Altogether, this describes a self-consistent, semiclassical evolution of the coupled light-matter system, where the matter component evolves quantum mechanically, while the part describing the electromagnetic field undergoes classical evolution and is driven by both the input field as well as by the collective molecular polarization feeding back into the cavity field. While Eq. (3) assumes a simplified, single-mode description of the electromagnetic field, we discuss the straightforward extension to multimode cavities in the Supplemental Material (SM) [49].

Perturbative expansion—While the above set of mean-field equations can be readily solved numerically, there is generally no closed analytical solution due to the nonlinearity of the molecular transitions. To gain more insight into the structure of the nonlinear response, we therefore proceed, akin to standard nonlinear spectroscopy, with a perturbative expansion of both the cavity field and matter density matrix in terms of the input field amplitude, i.e.,

$$\alpha(t) = \sum_{n=0}^{\infty} \eta^n \alpha^{(n)}(t), \quad \rho(t) = \sum_{n=0}^{\infty} \eta^n \rho^{(n)}(t). \quad (4)$$

From this, also other perturbative quantities can be derived such as, e.g., the n th order polarization $P^{(n)}(t) = \text{Tr}[\hat{\mu}\rho^{(n)}(t)]$ [53].

Furthermore, for both numerical and analytical considerations, it is advantageous to convert the master equation for the density matrix onto matrix-vector form, i.e., map the density matrix ρ onto a vector $\vec{\rho}$ and the superoperator acting on the density matrix onto a matrix (also referred to

as Liouville space) (see SM [49] for details on vectorization) [26,54,55]. Then, one obtains a closed system for the cavity field amplitude and the vectorized density matrix up to n th order

$$\dot{\alpha}^{(n)} = -\left(\frac{\kappa}{2} + i\omega_c\right)\alpha^{(n)} - i\mathcal{N}E_0P^{(n)} - \delta_{1n}f(t)e^{-i\omega_\epsilon t}, \quad (5a)$$

$$\dot{\vec{\rho}}^{(n)} = -i\mathcal{L}_0\vec{\rho}^{(n)} - iE_0\sum_{j=0}^n\left(\alpha^{(n-j)} + \alpha^{(n-j)*}\right)\mathcal{L}_\mu\vec{\rho}^{(j)}, \quad (5b)$$

in terms of the vectorized commutators $[\mathcal{H}_0, \rho^{(n)}] \rightarrow \mathcal{L}_0\vec{\rho}^{(n)}$ and $[\hat{\mu}, \rho^{(j)}] \rightarrow \mathcal{L}_\mu\vec{\rho}^{(j)}$, where $\mathcal{L}_0, \mathcal{L}_\mu$ are the free molecular and interaction Liouvillians, respectively. In particular, Eqs. (5a) and (5b) indicate that only the first-order cavity field is driven directly by the input pulse while at higher orders, the molecular density matrix is driven by lower orders of the cavity field. Importantly, Eq. (5b) shows that the (cavity) field driving the molecular density matrix can have nonlinear contributions in the input field, as opposed to free-space spectroscopy where the interaction Liouvillian driving the evolution of the density matrix is always linear in the (laser) input field [see Fig. 1(b)] [26]. In other words, the electric field experienced by the molecules is not the laser field interacting with the cavity from outside, but a distorted version of it due to the evolving material polarization inside the cavity. While for conventional spectroscopy with well-separated pulses, only a single multidimensional integral is obtained as solution for the n th order density matrix, many additional integrals arise in the cavity scenario ($2^{n-1} - 1$, see diagram in Fig. 2). Also, a straightforward analysis of Eqs. (5) reveals that the first order ($n = 1$) recovers the standard linear polariton response, characterized by a Rabi splitting proportional to $\sqrt{\mathcal{N}}$, arising from the hybridization of the cavity with the collective molecular bright mode [56–61].

Differential transmission—We now consider two pulses driving the cavity (pump p and probe p'), with input fields proportional to $\eta_j f_j(t - \tau_j)e^{-i\omega_j t}$ where $\eta_j, f_j, \omega_j, \tau_j$ for $j \in \{p, p'\}$ denote the amplitude, envelope, carrier frequency, and arrival time of the pump and probe pulses, respectively. We assume a time delay between the pump and probe of $\tau_\Delta = \tau_{p'} - \tau_p \geq 0$ [see sketch in Fig. 1(c)]. We proceed with a perturbative expansion in both pump (p) and probe (p') amplitudes (this can be generalized to arbitrary many pulses)

$$\alpha(t) = \sum_{n,m=0}^{\infty} \eta_p^n \eta_{p'}^m \alpha^{(n)(m)}(t), \quad (6a)$$

$$\rho(t) = \sum_{n,m=0}^{\infty} \eta_p^n \eta_{p'}^m \rho^{(n)(m)}(t), \quad (6b)$$

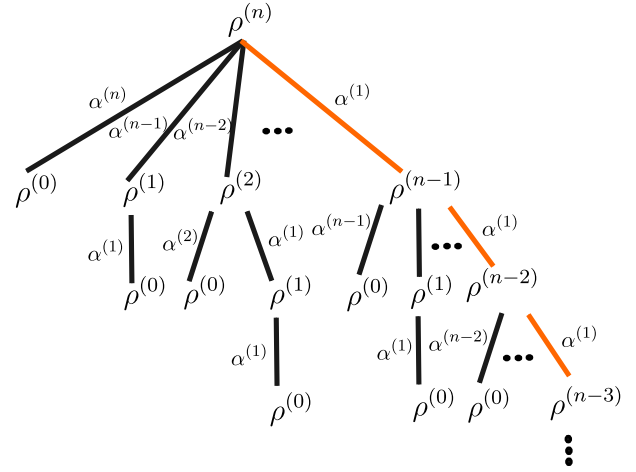


FIG. 2. Density matrix evolution. Graph illustrating the possible integrals contributing to the dynamical evolution of the n th order molecular density matrix $\rho^{(n)}(t)$. The orange path on the right illustrates the integral that is obtained for conventional free-space spectroscopy, corresponding to a sequence of first-order interactions that are linear in the input. The feedback of the nonlinear polarization onto the cavity field adds $2^{n-1} - 1$ other possible pathways.

where the first and second indices refer to the orders in the pump and probe fields, respectively. This allows us to derive a perturbative set of equations for $\alpha^{(n)(m)}, \vec{\rho}^{(n)(m)}$ similar to Eqs. (5), now describing pump-probe spectroscopy (see SM [49] for details).

From the solution for the total cavity field, the cavity transmission and reflection can be obtained using standard input-output theory [62] (valid as long as the system is not in the ultrastrong coupling regime [63,64]). Since nonlinearities are typically weak, instead of absolute quantities, differential (nonlinear) corrections to the response are commonly considered. For the collinear geometry in question, we define the differential transmission (DT) spectrum by subtracting the (normalized) probe transmission without the pump from the probe transmission in the presence of the pump field

$$\Delta T(\omega) = T_{p'}^{\text{pump on}}(\omega) - T_{p'}^{\text{pump off}}(\omega), \quad (7)$$

where the transmission is related to the intracavity field as $T_{p'}(\omega) = |(\kappa/2)\alpha(\omega)/[\eta_{p'}f_{p'}(\omega)]|^2$. The contribution $T_{p'}^{\text{pump on}}(\omega)$ depends on the time delay between the pulses τ_Δ . By considering only the lowest (third) order correction due to the pump and probe, the DT may be expressed as (see SM for derivation [49])

$$\Delta T(\omega) \approx \left(\frac{\kappa}{2}\right)^2 \eta_p^2 \frac{2\text{Re}[\alpha^{(0)(1)*}(\omega)\alpha^{(2)(1)}(\omega)]}{f_{p'}(\omega)^2}, \quad (8)$$

describing interference [four-wave mixing (4WM)] between the probe (acting as local oscillator) and the

third-order cavity field $\alpha^{(2)(1)}$. Beware that Eq. (8) is not the usual pump-probe signal; instead, it contains different phase contributions (see paragraph below). The DT is plotted as a function of delay time in Figs. 3(a) and 3(b) for two-level systems (2LSs) with transition dipole moment μ_{ge} coupled to the cavity at strength $g = \mu_{ge}E_0$, for two different degrees of dephasing. The plots in Figs. 3(c) and 3(d) show the dynamics of the polarization and population created by an initial pump pulse which are then probed after a time τ_Δ . At early times, oscillatory behavior around the polariton frequencies is observed, corresponding to coherences between the upper and lower polariton. Dephasing leads to a faster decay of coherence and the creation of a stationary population in the molecules, even after the cavity photon has decayed. In the long-time limit, this manifests as a contracted Rabi splitting, characterized by a DT as shown by the black cross-section curve in Fig. 3(b) [20,23,33].

Separating excitation pathways—The nonlinear signal stemming from the n th order density matrix generally contains many possible excitation pathways. For instance, even for free-space spectroscopy of a 2LS, there are eight Feynman diagrams contributing to the third-order nonlinear response [26]. In experiments, it is therefore highly desired

to isolate certain excitation pathways and thereby understand the different contributions to the nonlinear signal, which here, owing to the collinear geometry in question, can be achieved by phase cycling. This method is equivalent to the more standard phase matching procedure with oblique pulses, which, however obfuscates our formalism as it requires the description of a multimode cavity [28,29,65–68].

To this end, we will consider the effect of additional phases applied to the pump (Φ_p) and probe ($\Phi_{p'}$) pulses. These phases are subsequently also imprinted onto the cavity field and the molecular density matrix, enabling us to disentangle different excitation pathways characterized by specific phase sequences [26,28]. The cavity field and density matrix are now expanded as

$$\alpha(t) = \sum_{n,m,\vec{v}} \eta_p^n \eta_{p'}^m e^{-i\vec{v} \cdot \vec{\Phi}} \alpha_{\vec{v}}^{(n)(m)}(t), \quad (9a)$$

$$\rho(t) = \sum_{n,m,\vec{v}} \eta_p^n \eta_{p'}^m e^{-i\vec{v} \cdot \vec{\Phi}} \rho_{\vec{v}}^{(n)(m)}(t), \quad (9b)$$

where $\vec{\Phi} = (\Phi_p, \Phi_{p'})^\top$ denotes the vector of pump and probe phases, and $\vec{v} = (v_p, v_{p'})^\top$ with $v_p, v_{p'} \in \mathbb{Z}_0$ denotes the phase coefficients which also depend on the pump and probe orders $v_p(n), v_{p'}(m)$ (see SM [49]). The raw DT signal from Eq. (8) can then be separated into different phase contributions

$$\Delta T_{\vec{v}}(\omega) \approx \left(\frac{\kappa}{2}\right)^2 \eta_p^2 \frac{2\text{Re}[\alpha_{(0,-1)}^{(0)(1)*}(\omega) \alpha_{\vec{v}}^{(2)(1)}(\omega)]}{f_{p'}(\omega)^2}, \quad (10)$$

where $(0, -1)$ is the only phase contribution for the conjugate probe field [49] and the total signal is obtained as $\Delta T = \sum_{\vec{v}} \Delta T_{\vec{v}}$. Two phase combinations contribute to $\alpha^{(2)(1)}$: $(0, 1)$ and $(2, -1)$ [along with the complex conjugates $(0, -1)$ and $(-2, 1)$], corresponding to pump-probe and double-quantum coherence (DQC) spectroscopies, respectively [Figs. 4(a) and 4(b)]. The pump-probe pathway $(0, 1) \equiv \pm\Phi_p \mp \Phi_p + \Phi_{p'}$ creates populations and zeroth-order quantum coherences via two pump pulses of opposite phase [Fig. 4(c)]. The DQC pathway $(2, -1) \equiv 2\Phi_p - \Phi_{p'}$ is negligible outside the cavity because it requires the two-level system to be excited twice. However, inside the cavity, the probe can act on the pump-induced coherence within the lifetime of the bright state $(\kappa + \gamma_\phi)^{-1}$ [left diagram in Fig. 4(d)], enabling this pathway to arise within this time interval. The right diagram in Fig. 4(d) does not contribute to the probe DT since the pump has not acted yet. Consequently, the $(2, -1)$ pathway isolates the bright state only, in contrast to the $(0, 1)$ pathway, which accesses both bright and dark states (and probes only dephased dark states in the long time

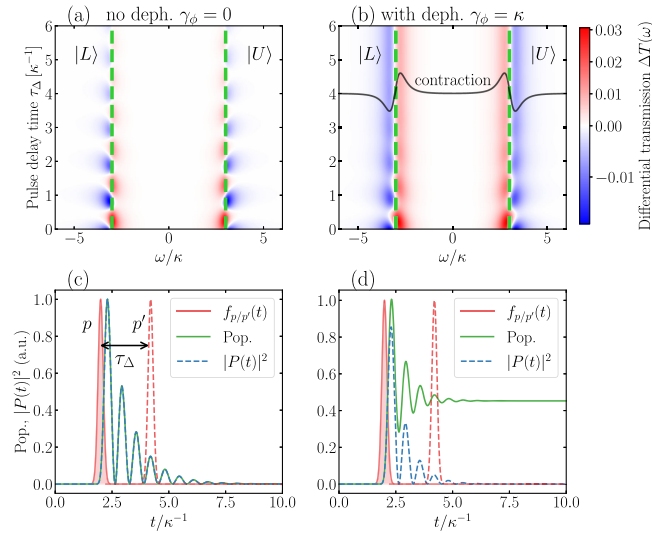


FIG. 3. Differential transmission. DT spectra $\Delta T(\omega)$ for a 2LS with resonance frequency ω_0 as a function of the delay time between the pump and probe pulse τ_Δ for (a) weak ($\gamma_\phi = 0$), and (b) strong ($\gamma_\phi = \kappa$) dephasing of the transition. The green dashed lines show the location of the (linear response) polariton frequencies. The black curve in (b) shows a cross section through the DT at $\tau_\Delta = 4\kappa^{-1}$. The results are plotted in a frame rotating at the central pulse frequency. Panels (c) and (d) depict the dynamics of the population and polarization $|P(t)|^2$ created by the initial pump pulse arriving at $\tau_p = 2\kappa^{-1}$ which are then probed by the second pulse at delay τ_Δ . The 2LS is initialized in the ground state, with parameters $g\sqrt{N} = 3\kappa$, $\tau_w = 0.05\kappa^{-1}$, $\omega_0 = \omega_c$, and carrier frequency $\omega_p = \omega_{p'} = \omega_0$ for both pulses.

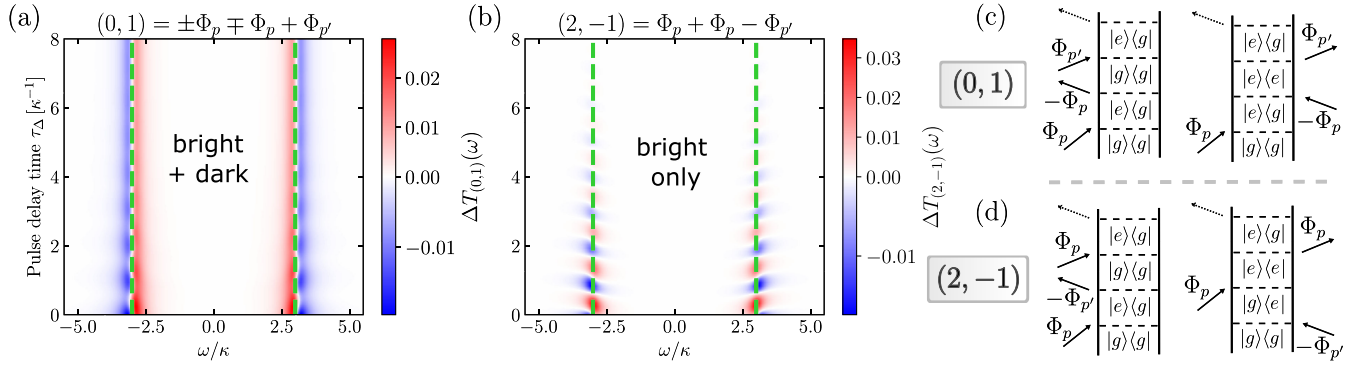


FIG. 4. Separating phase contributions to the nonlinear polariton response. (a), (b) Plot of phase contributions $(0,1)$ and $(2,-1)$ to $\Delta T(\omega)$ as a function of the delay time between the pulses. The $(2,-1)$ contribution decays on a time scale of $\sim(\kappa + \gamma_\phi)^{-1}$, as it relies on the probe acting on the coherence created by the pump, therefore probing the bright state only. The green dashed lines show the (linear response) polariton frequencies. We chose $g\sqrt{N} = 3\kappa$, $\gamma_\phi = 0.1\kappa$, $\omega_p = \omega_{p'} = \omega_0$, and a temporal width of $\tau_w = 0.1\kappa^{-1}$ for both pulses. The double-sided Feynman diagrams in (c) and (d) represent processes contributing to the different phase combinations of the third-order cavity field $\alpha^{(2)(1)}$. The dotted arrows indicate the decay of the coherence.

limit). Equation (10) satisfies the phase matching conditions for degenerate 4WM with two pulses [28].

Discussion and conclusions—We have presented an efficient and broadly applicable framework for nonlinear polariton spectroscopy, based on a mean-field evolution of the coupled light-matter system. The formalism extends conventional free-space spectroscopy by accounting for the feedback of the molecular polarization onto the cavity field and allows for an understanding of the polariton response beyond the stationary limit [33]. Expanding the cavity field and molecular density matrix in terms of the input fields' amplitudes and phases provides a systematic understanding of the various contributions to the nonlinear polariton response, and allows us to separate purely bright state contributions from dark reservoir states.

The formalism offers potential for extension and generalization via multiple avenues. For instance, while our current treatment relies on a semiclassical mean-field description, valuable extensions could involve capturing nonclassical photon statistics or many-body correlations among particles, e.g., by using cumulant or cluster expansion techniques [69,70]. Other areas of interest are the study of spatially resolved nonlinear polariton spectroscopy, which would involve generalizing the zero-dimensional cavity Eq. (3) to the 3D set of cavity modes afforded by classical electromagnetic simulations [71–75], or cavities exhibiting inherent nonlinearities [76]. Furthermore, while we have focused on pump-probe spectroscopy involving two pulses, the formalism can be readily applied to 2D or multidimensional coherent spectroscopy, which may also help to disentangle pure electronic dephasing from vibronic relaxation channels. Finally, let us emphasize that our approach is not limited to molecular polaritons, but could also be extended to other strongly coupled light-matter platforms, such as exciton polaritons in inorganic semiconductor microcavities [77] or atomic ensembles.

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